

1. **(Gaussian Elimination)** Calculate the determinant $|A|$ using partial pivoting. Fill in the blank values at each step. If a vector or matrix doesn't change from one step to the next, you don't have to fill it in (just mark it as the same). Some fields may be unused. ($\mathbf{l}$ is used to record the positions/indices of rows)

$$|A| = \begin{vmatrix} 0 & 1 & 5 & 7 \\ 0 & 2 & 4 & 2 \\ 1 & 3 & 2 & 2 \\ 0 & 1 & 3 & 5 \end{vmatrix} = (-1) * 1 * 2 * 3 * 2 = -12$$

$$\begin{bmatrix} 0 & 1 & 5 & 7 \\ 0 & 2 & 4 & 2 \\ 1 & 3 & 2 & 2 \\ 0 & 1 & 3 & 5 \end{bmatrix} \quad \mathbf{l} = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 3 & 6 \\ 0 & 2 & 4 & 2 \\ 1 & 3 & 2 & 2 \\ 0 & 0 & 1 & 4 \end{bmatrix} \quad \mathbf{l} = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 3 & 6 \\ 0 & 2 & 4 & 2 \\ 1 & 3 & 2 & 2 \\ 0 & 0 & 0 & 2 \end{bmatrix} \quad \mathbf{l} = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 4 \end{bmatrix}$$

Then answer the following questions:

- If we denote the elimination as $\mathbf{PA} = \mathbf{LU}$, please write down the $\mathbf{P}$, $\mathbf{L}$, and $\mathbf{U}$. Also verify the equation $\mathbf{PA} = \mathbf{LU}$ to double check your work.

$$\mathbf{P} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \mathbf{L} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & \frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{3} & 1 \end{bmatrix}, \mathbf{U} = \begin{bmatrix} 1 & 3 & 2 & 2 \\ 0 & 2 & 4 & 2 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & 0 & 2 \end{bmatrix} \mathbf{e}$$

- What is $|A^{-1}|$? What formula do you use to calculate this?

$$|A^{-1}| = \frac{1}{|A|} = \frac{1}{-1 * 2 * 3 * 2} = -\frac{1}{12}$$

2. **(LU Decomposition)**

- Calculate the missing column of the matrix inverse A^{-1} given the LU decomposition of a matrix $A = \mathbf{LU}$.
- Estimate the condition number for A using the $\|\cdot\|_1$ and $\|\cdot\|_\infty$ norms.

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 1 & 2 \end{bmatrix}, \quad U = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}, \text{ and}$$

$$A^{-1} = \begin{bmatrix} \frac{11}{6} & -\frac{7}{24} & \\ -\frac{2}{3} & \frac{1}{12} & \\ \frac{1}{2} & \frac{1}{8} & \end{bmatrix},$$

Since $LUA^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, we have $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} \frac{11}{6} & -\frac{7}{24} & \\ -\frac{2}{3} & \frac{1}{12} & \\ \frac{1}{2} & \frac{1}{8} & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$

then $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$

Define $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$, we have $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$

Use substitution, we can find $\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$, then $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$

Use back substitution, we have $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{12} \\ \frac{1}{6} \\ -\frac{1}{4} \end{bmatrix}$

We know $\kappa(A) = \|A\| \|A^{-1}\|$, and $\|A\| \approx \sqrt{\|A\|_1 \|A\|_\infty}$ (also $\|A^{-1}\| \approx \sqrt{\|A^{-1}\|_1 \|A^{-1}\|_\infty}$).

First, $A = LU = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 10 & 6 \\ 3 & 9 & 1 \end{bmatrix}$, we can have $\|A\|_1 = \max(6, 21, 8) = 21$ and $\|A\|_\infty =$

$\max(4, 18, 13) = 18$. Thus $\|A\| \approx 19.4$.

Then, $A^{-1} = \begin{bmatrix} \frac{11}{6} & -\frac{7}{24} & -\frac{1}{12} \\ -\frac{2}{3} & \frac{1}{12} & \frac{1}{6} \\ \frac{1}{2} & \frac{1}{8} & -\frac{1}{4} \end{bmatrix}$, we can have $\|A^{-1}\|_1 = \max(3, \frac{1}{2}, \frac{1}{2}) = 3$ and $\|A\|_\infty =$

$\max(\frac{53}{24}, \frac{11}{12}, \frac{7}{8}) = \frac{53}{24}$. Then $\|A^{-1}\| \approx 2.57$.

$$\kappa(A) = \|A\| \|A^{-1}\| \leq 19.4 * 2.57 = 49.9$$

3. (Roots of equations)

- 1) Graphically demonstrate the False position method on the attached function fig for 3 iterations given the interval $[-2, 4]$. Draw the line representing each

c_n . Label the line with (1) the iteration n , (2) the x-axis position, and (3) the interval side (a or b .) that c_n replaces. Please also record the procedures in the following format.

	a	b	$f(a)$	$f(b)$	c	$f(c)$
#1						
#2						
#3						

- 2) Graphically demonstrate the combined Newton's and Bisection methods on the attached function fig for 3 iterations given the interval $[-2, 4]$. Try to use Newton's method as possible. Draw the line representing each c_n or x_n . Label the line with (1) the iteration n , (2) the x-axis position, (3) the interval side (a or b) that c_n replaces (if needed), and (4) the procedure to determine x_{n+1} . Please also record the procedures in the following formats.

If Bisection method is used

	a	b	$f(a)$	$f(b)$	c	$f(c)$
#1						
#2						

If Newton's method is used

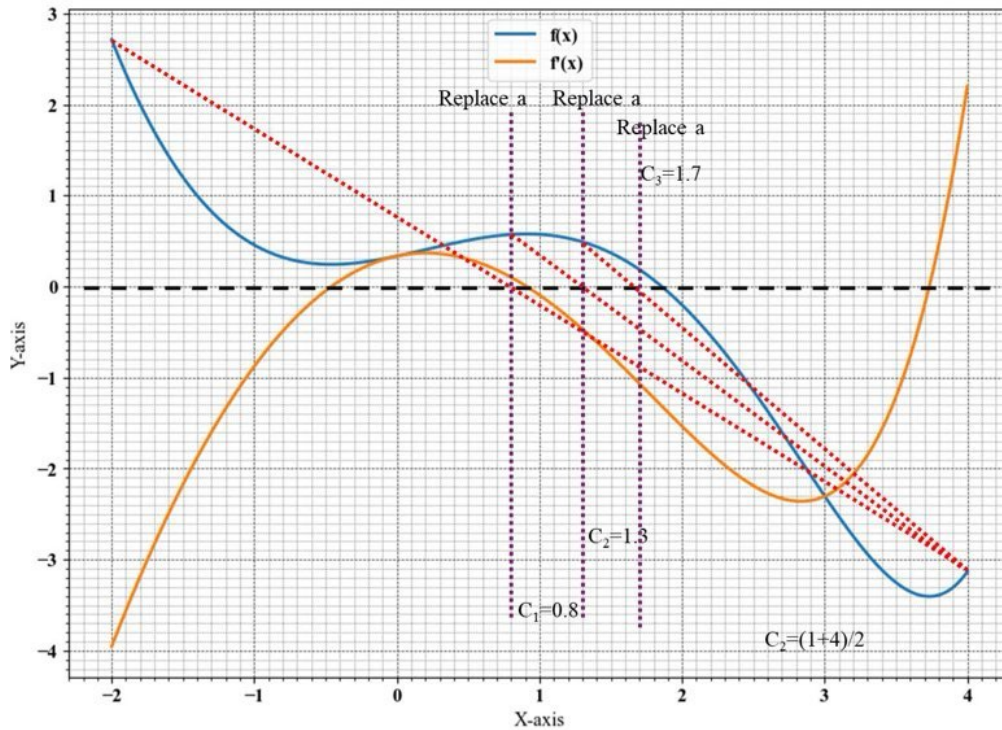
	x_n	$f(x_n)$	$f'(x_n)$	$\frac{f(x_n)}{f'(x_n)}$	x_{n+1}
#1					
#2					

The two function figs are the same. Use one for False position method and another one for Newton's/Bisection method.

1) False Position

We can determine the position of c using $c = \frac{af(b)-bf(a)}{f(b)-f(a)}$, or the x-intercept for the line $\frac{f(a)f(b)}{f(a)f(b)}$, as shown in the following.

	a	b	$f(a)$	$f(b)$	c	$f(c)$
#1	-2	4	2.7	-3.1	0.8	0.6
#2	0.8	4	0.6	-3.1	1.3	0.5
#3	1.3	4	0.5	-3.1	1.7	0.2



2) Combined Newton's and Bisection methods

We first try the Newton's method. Since the interval is $[-2, 4]$, we have $x_1 = 1$. Fill the table as follows

	x_n	$f(x_n)$	$f'(x_n)$	$\frac{f(x_n)}{f'(x_n)}$	x_{n+1}
#1	1	0.6	-0.1	-6	7

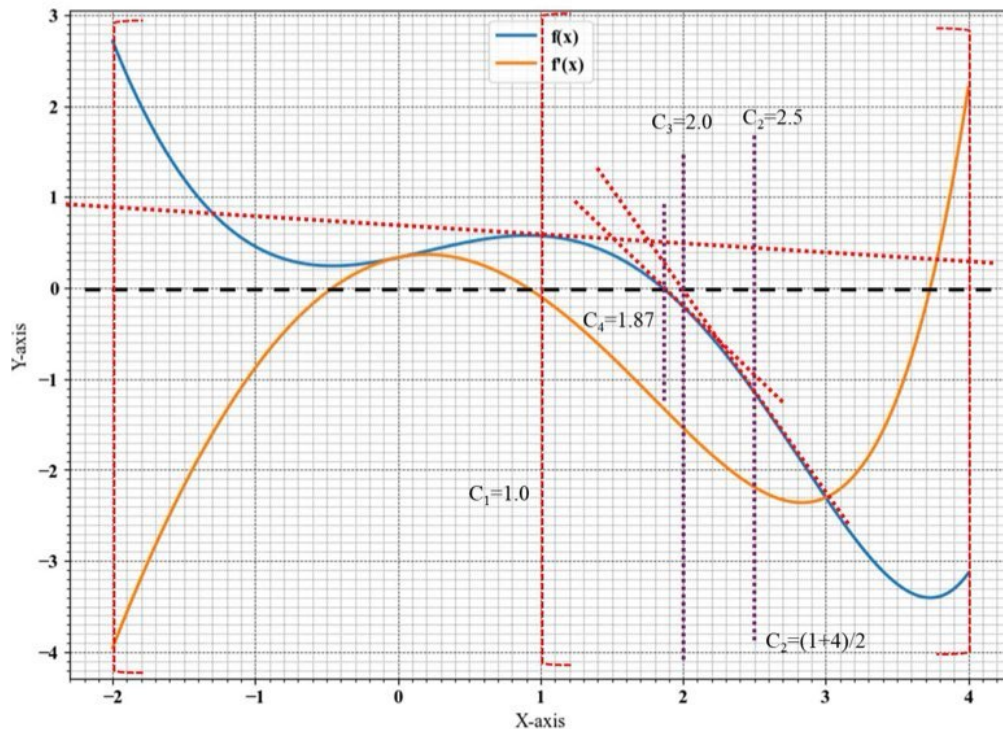
We can also find this on the image. The tangent line doesn't intersect the x-axis. Thus we try the bisection method

	a	b	$f(a)$	$f(b)$	c	$f(c)$
#1	-2	4	2.7	-3.1	1	0.6
#2	1	4	0.6	-3.1	2.5	1.15

Use the Newton's method again, we use the $c = 2.5$ obtained from the bisection method as the new start point

	x_n	$f(x_n)$	$f'(x_n)$	$\frac{f(x_n)}{f'(x_n)}$	x_{n+1}
#1	2.5	-1.15	-2.2	- 6	2.0
#2	2.0	-0.2	-1.55	- 0.13	1.87

It's already very close to 0, we can just stop here.



4. (**Horner's Algorithm**) Calculate $f(x) = \frac{\log(x)}{\sqrt{x}}$ for $x = 1.2$ using Horner's algorithm to an error of $O(x^4)$ and 3 digits of precision. Demonstrate Horner's algorithm using synthetic division. Show your work, including all derivations.

Expand $\log(x)$ at $x = 1$, we have $\log(x) = (x - 1) - \frac{1}{2}(x - 1)^2 + \frac{1}{3}(x - 1)^3$

Expand $\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$ at $x = 1$, we have $x^{-\frac{1}{2}} = 1 - \frac{1}{2}(x - 1) + \frac{3}{8}(x - 1)^2 - \frac{5}{16}(x - 1)^3$

Then we can expand $f(x) = \frac{\log(x)}{\sqrt{x}}$ to the order of $O(x^3)$ as

$$f(x) = \frac{\log(x)}{\sqrt{x}} = (x-1) - (x-1)^2 + \frac{23}{24}(x-1)^3 + O(x^4)$$

We have $f(x) \approx 0.958x'^3 - x'^2 + x' + 0$ if define $x' = x - 1$

Using synthetic division, we have

$$\begin{array}{r|rrrr}
 0.2 & 0.958 & -1 & 1 & 0 \\
 & 0.192 & -0.162 & 0.168 & \\
 \hline
 & 0.958 & -0.808 & 0.838 & 0.168
 \end{array}$$

5. (Random number generators)

- Assume a linear congruential generator with the following properties: modulus $= 2^5$, increment $= 3$, multiplier $= 5$. Calculate the first 5 values using a seed $x_0 = 22$. Show the equations defining the LCG.

$$(5 * 22 + 3) \% 2^5 = 17$$

$$(5 * 17 + 3) \% 2^5 = 24$$

$$(5 * 24 + 3) \% 2^5 = 27$$

$$(5 * 27 + 3) \% 2^5 = 10$$

$$(5 * 10 + 3) \% 2^5 = 21$$

- Assume the modulus $m = 12$, please select a multiplier (a), increment (c) and seed x_0 according to the Hull-Dobell Theorem to maximize the period of the linear congruential generator. (**Hint:** the a can be larger than m)

The Hull-Dobell Theorem conditions are:

- c and m are coprime ($\gcd(c, m) = 1$).
- $a - 1$ is divisible by all prime factors of m
- $a - 1$ is a multiple of 4 if m is a multiple of 4

Referring Hull-Dobell theorem, $a - 1$ is a multiple of 4 since $m = 12$ is a multiple of 4. Since 4 and 8 are both not divisible by 3, we choose $a = 12 + 1$. c can be 5, 7, 11...

- Calculate the sequence to verify your design.

Let's set $a = 13$, $m = 12$, $c = 5$, and $x_0 = 1$

$$(13 * 1 + 5) \% 12 = 6 = 0b0110$$

$$(13 * 6 + 5) \% 12 = 11 = 0b1011$$

$$(13 * 11 + 5) \% 12 = 4 = 0b0100$$

$$(13 * 4 + 5) \% 12 = 9 = 0b1001$$

$$(13 * 9 + 5) \% 12 = 2 = 0b0010$$

$$(13 * 2 + 5) \% 12 = 7 = 0b0111$$

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(13 * 1 + 5)%12 = 0 = 0b0000  
(13 * 1 + 5)%12 = 5 = 0b0101  
(13 * 1 + 5)%12 = 10 = 0b1010  
(13 * 1 + 5)%12 = 3 = 0b0011  
(13 * 1 + 5)%12 = 8 = 0b1000  
(13 * 1 + 5)%12 = 1 = 0b0001
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- What's the problem with this sequence in terms of "randomness".

If we check the binary representation, we can easily find that the last bits have a pattern of "00,01,10,11..." The sequence is not so random.